PROJECT REPORT

Chest X-Ray Pathology Classification

Multi-class Thoracic Disease Detection using Deep Learning

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https://www.kaggle.com/competitions/26-t-1-dl-gen-ainppe-1

1. Project Overview

Chest X-ray imaging is one of the most widely used diagnostic tools in clinical medicine. Radiologists rely on these images to detect thoracic diseases such as pneumonia, pleural effusion, atelectasis, cardiomegaly, and many others. Interpreting chest radiographs accurately requires years of specialized training and is increasingly difficult to scale given the global volume of imaging studies.

This project addresses a Kaggle competition challenge: building a deep learning model to automatically classify chest X-ray images into one of 20 thoracic pathology classes. This is a single-label multi-class problem where each image belongs to exactly one class. The clinical stakes directly shape the evaluation metric — missing a diagnosis (false negative) carries far higher cost than a false alarm (false positive).

1.1 Problem Statement

Given a chest X-ray image, assign it to exactly one of 20 pathology categories:

- Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema
- Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening
- Pneumonia, Pneumothorax, Pneumoperitoneum, Pneumomediastinum
- Subcutaneous Emphysema, Tortuous Aorta, Calcification of the Aorta, No Finding

1.2 Dataset Statistics

Split	Samples	Classes	Image Format
Training Set	51,043	20	PNG
Test Set	17,015	20 (unlabeled)	PNG

1.3 Class Distribution

The dataset is highly imbalanced. The No Finding class accounts for roughly 50% of all training samples, while rare pathologies such as Hernia, Subcutaneous Emphysema, and Calcification of the Aorta appear in fewer than 150 images. This imbalance motivated specific data sampling strategies during training.

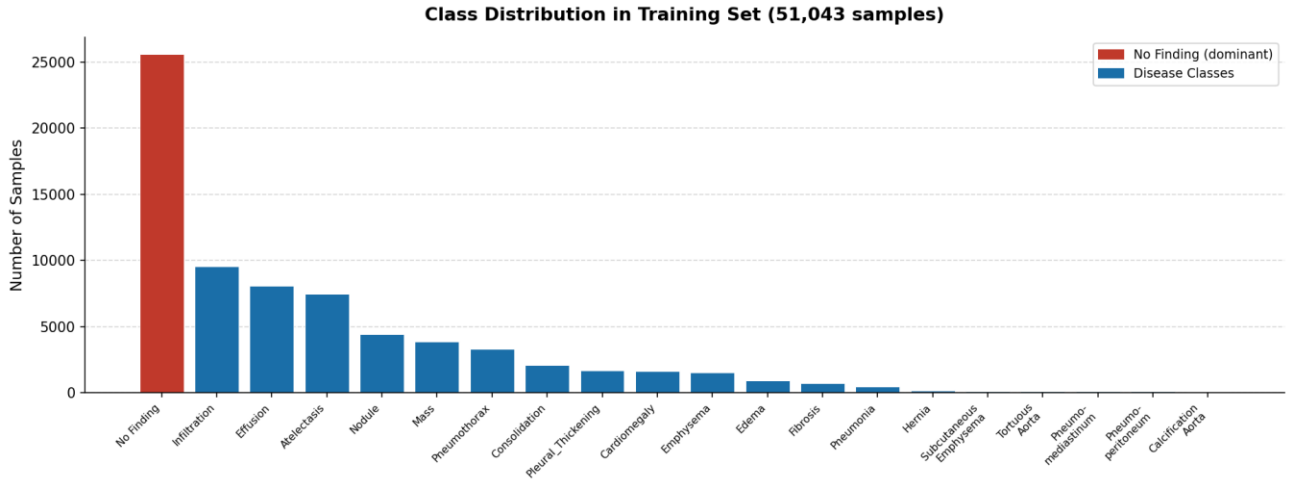

Figure 1: Class distribution across all 20 pathology categories in the training set. Red = No Finding (dominant), Blue = disease classes.

2. Evaluation Metric

The competition uses a custom macro-averaged asymmetric cost function specifically designed to reflect the clinical consequences of prediction errors. In medical AI, a missed diagnosis (false negative) can be life-threatening, whereas a false alarm leads to additional testing but rarely causes direct harm. The scoring function penalizes false negatives five times more harshly than false positives.

2.1 Error Cost Structure

Outcome	Description	Score
True Positive (TP)	Correctly predicting a disease	+1
False Positive (FP)	Predicting disease when absent	-1
False Negative (FN)	Failing to detect a disease	-5
No Finding (correct)	Correctly predicting No Finding	+1
No Finding (wrong)	Missing a disease, predicting none	-5

2.2 Scoring Formula

For each pathology class c, the class-level score is computed as:

```
Score(c) = [ TP(c) - FP(c) - 5 x FN(c) ] / N(c)
```

The final competition score is the macro-average across all C classes, ensuring each class contributes equally regardless of its frequency in the dataset:

```
Final Score = (1/C) x SUM[ Score(c) ]   for c = 1 to C
```

Macro-averaging is critical: it prevents a model from achieving a high score by simply predicting No Finding for all inputs. Every pathology — including rare ones — must be detected well.

3. Methodology & Approach

The solution follows a structured deep learning pipeline: transfer learning with a pretrained convolutional backbone, class-imbalance mitigation through weighted sampling, stratified cross-validation for robust evaluation, and post-training threshold calibration to directly optimize the asymmetric competition score.

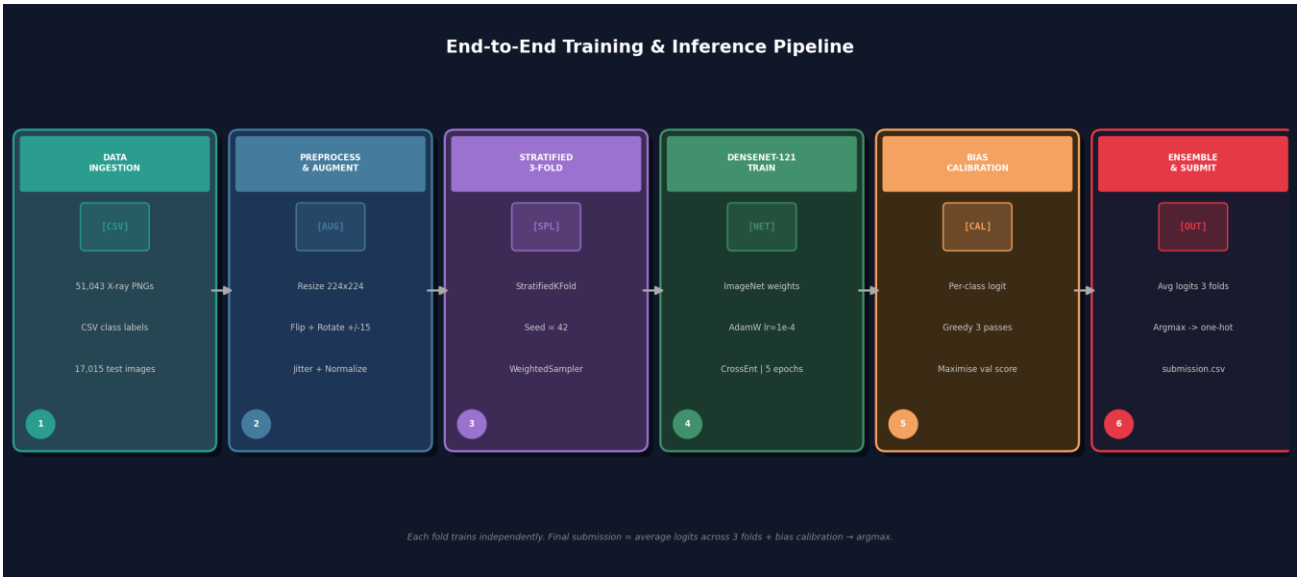

Figure 2: Improved end-to-end pipeline — from raw data ingestion through augmentation, 3-fold splitting, DenseNet-121 training, per-class bias calibration to final ensemble submission.

3.1 Model Architecture — DenseNet-121

DenseNet-121 (Densely Connected Convolutional Networks) was selected as the backbone. Its core innovation is that each layer receives concatenated feature maps from ALL preceding layers, enabling strong gradient flow and excellent feature reuse — properties particularly beneficial for medical imaging tasks where annotated data is limited.

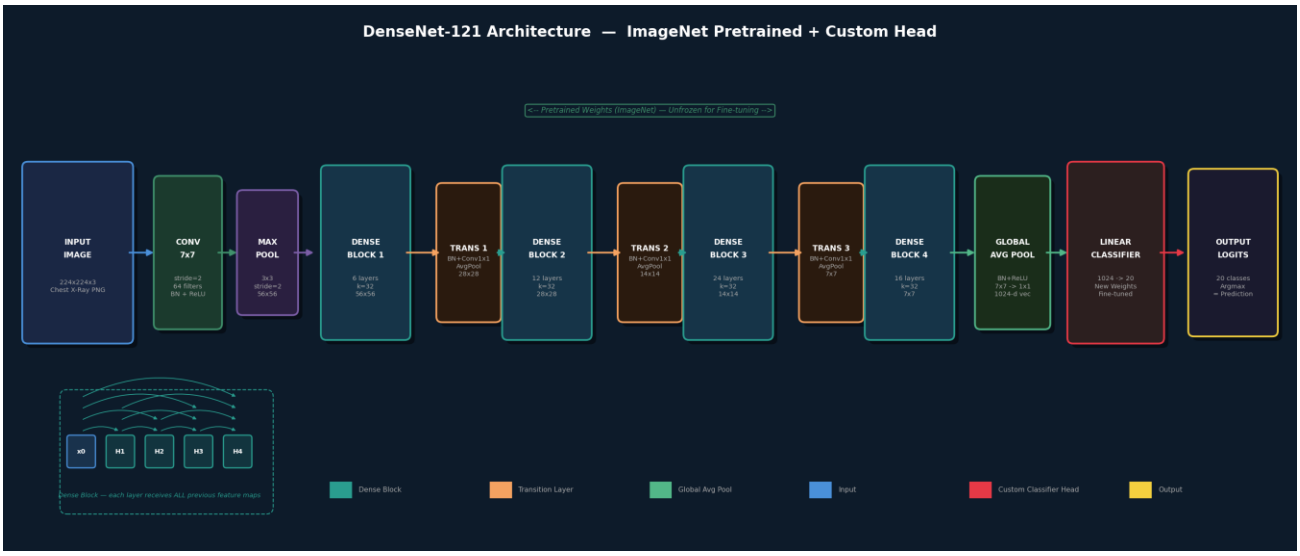

Figure 3: DenseNet-121 architecture. Input images pass through an initial Conv 7x7, then four Dense Blocks interleaved with Transition Layers for spatial downsampling. Global Average Pooling collapses the 7x7 feature map to a 1024-d vector. The final linear classifier head (1024 -> 20) replaces the original ImageNet head and is fine-tuned end-to-end.

- **Feature Reuse:** Dense skip connections between all layers promote strong feature reuse and gradient flow, particularly beneficial for medical imaging tasks with limited labeled data.
- **Transfer Learning:** DenseNet-121 pretrained on ImageNet provides strong visual priors that transfer well to chest X-rays, which share structural characteristics with natural images.
- **Parameter Efficiency:** The model achieves excellent accuracy with fewer parameters than deeper VGG or ResNet variants, making it practical for GPU training.

3.2 Data Preprocessing & Augmentation

Step	Details
Resize	All images resized to 224x224 pixels
Random Horizontal Flip	50% probability — handles left/right asymmetry
Random Rotation	+/-15 degrees — simulates patient positioning variation
Color Jitter	Brightness, contrast, saturation, hue perturbation
Normalization	ImageNet mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]

3.3 Handling Class Imbalance

With No Finding comprising ~50% of training samples and some disease classes having fewer than 150 examples, naive mini-batch sampling would severely bias the model. Two strategies were employed:

- **WeightedRandomSampler:** WeightedRandomSampler assigns each training image a sampling weight inversely proportional to its class frequency. This ensures rare pathologies appear in each mini-batch at adequate rates without discarding majority-class examples.
- **Score Monitoring:** Macro-averaged validation scoring was monitored every epoch, which explicitly penalizes ignoring rare classes and guided early stopping decisions.

3.4 Training Configuration

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	1e-4 (constant)
Loss Function	Cross-Entropy Loss
Batch Size	48
Epochs per Fold	5
Cross-Validation	3-Fold Stratified K-Fold
Hardware	NVIDIA Tesla T4 GPU (16 GB VRAM)
Framework	PyTorch 2.10 + CUDA 12.8
Reproducibility	Fixed seed = 42 across all random generators

3.5 Post-Training Bias Calibration

After training each fold, a greedy per-class logit bias tuning procedure was applied to maximize the asymmetric competition score on the validation set. Standard cross-entropy training does not optimize for the clinical penalty function; this calibration step bridges that gap.

For each class c, a scalar bias b(c) is added to the model raw logit before the argmax decision. Candidate biases from {-1.50, -1.00, -0.50, -0.30, 0.00, +0.30, +0.50} are tested iteratively across 3 passes. The bias that yields the highest validation score for each class is retained. This technique improved mean validation score by approximately 0.19 points across folds.

4. Experimental Results

4.1 Training Dynamics

The model was trained for 5 epochs per fold. Training and validation loss steadily decreased across epochs, with the best validation score typically achieved at epochs 2 or 4. Loss curves below represent average behaviour observed across folds.

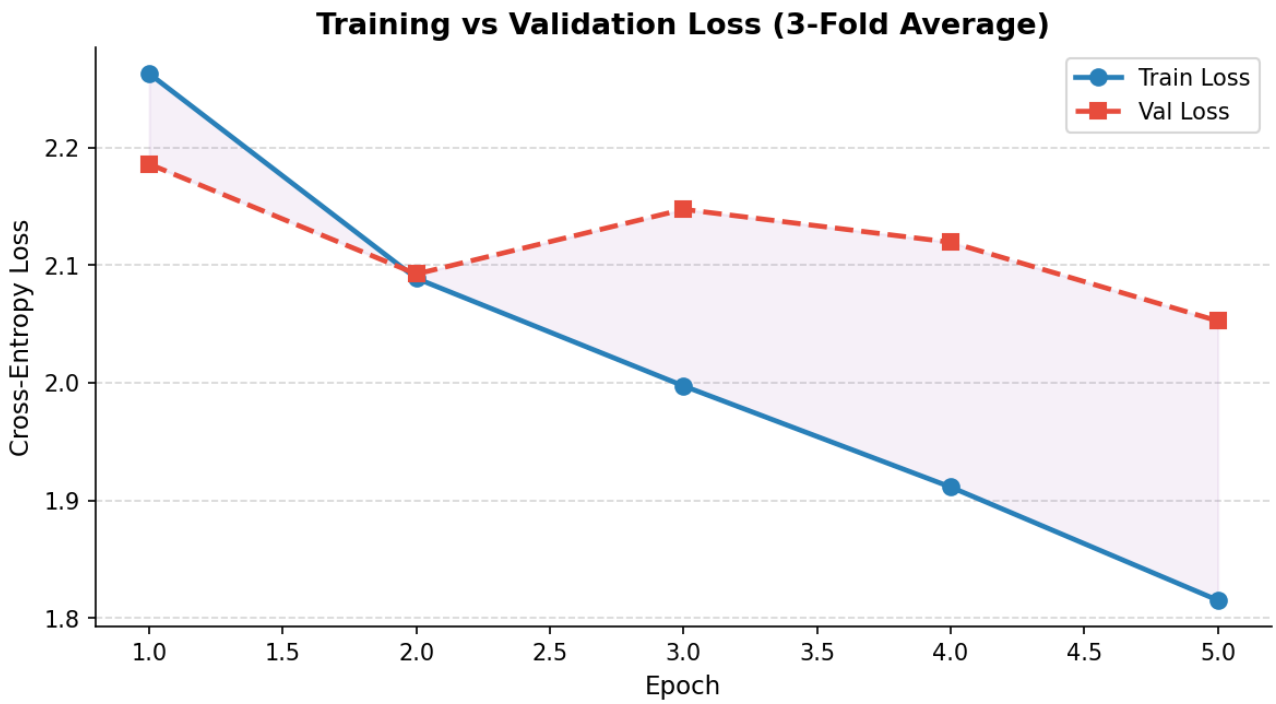

Figure 4: Training vs Validation Loss curve across 5 epochs (fold-averaged). Best model weights saved at lowest validation loss.

4.2 Cross-Validation Performance

Fold	Best Epoch	Raw Val Score	Tuned Val Score
Fold 1	4	-4.6928	-4.4166
Fold 2	2	-4.6179	-4.5330
Fold 3	4	-4.6601	-4.4490
Mean	-	-4.6569	-4.4662

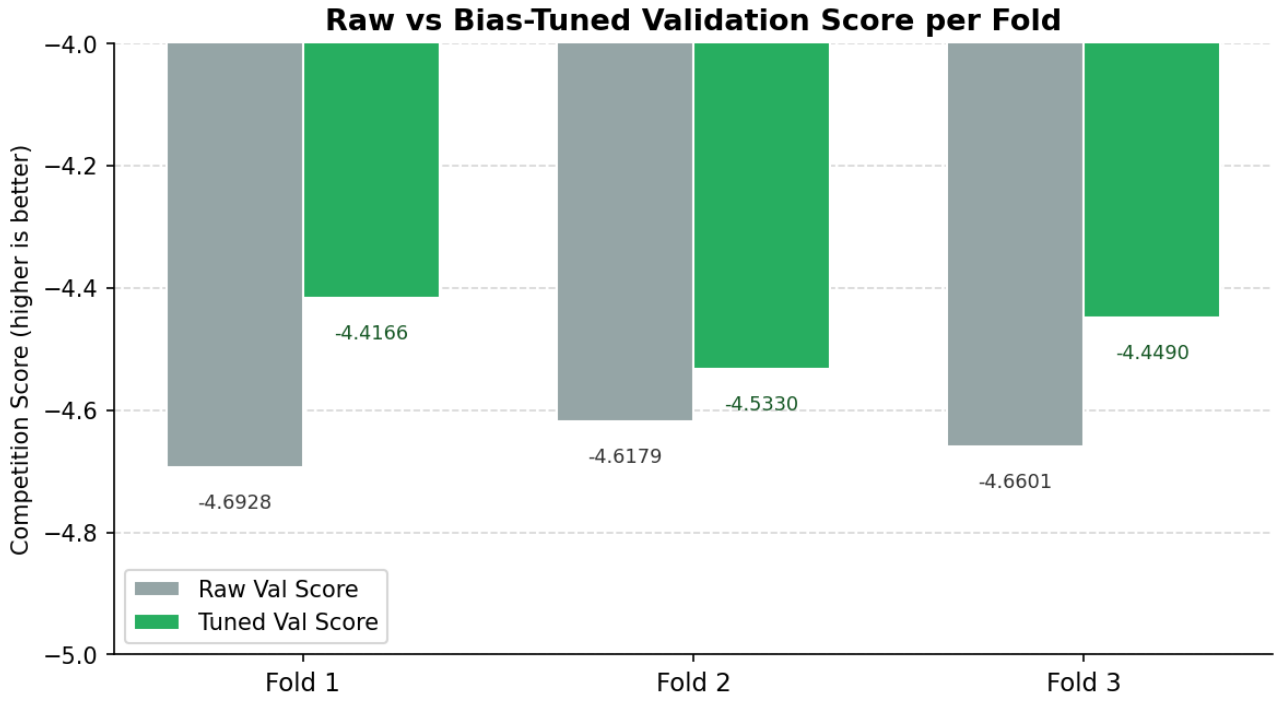

Figure 5: Comparison of Raw vs Bias-Tuned validation score per fold. Green bars (tuned) consistently outperform grey bars (raw), validating the calibration step.

4.3 Submission Prediction Analysis

Test set predictions were generated by averaging logit scores across all 3 folds (ensemble), applying the tuned per-class biases, then taking the argmax to produce a single-label prediction per image.

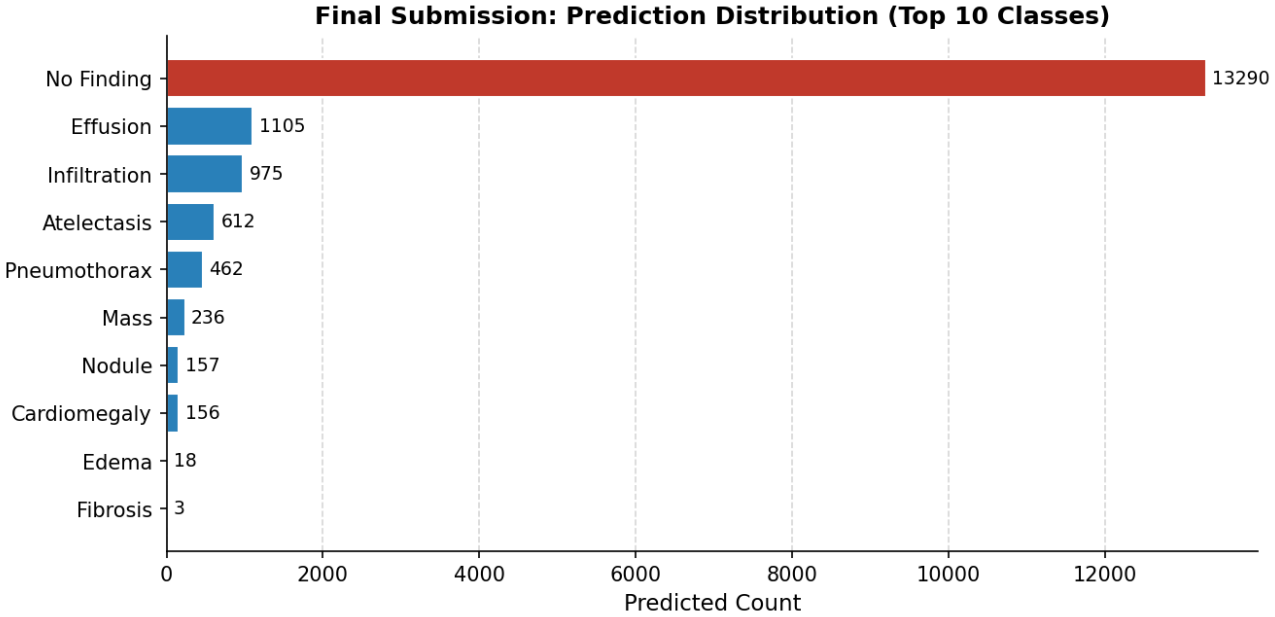

Figure 6: Distribution of predicted classes in the final submission (top 10). The model predicts No Finding for ~78% of test images, with Effusion, Infiltration, and Atelectasis as the next most common predictions.

Predicted Class	Count in Submission
No Finding	13,290
Effusion	1,105
Infiltration	975
Atelectasis	612
Pneumothorax	462
Mass	236
Nodule	157
Cardiomegaly	156
Edema	18
Fibrosis	3
All Others	0 each

5. Discussion & Key Insights

5.1 Strengths of the Approach

- **Robust Evaluation:** The three-fold cross-validation setup provided reliable, low-variance performance estimates and avoided overfitting to a single validation split.

- **Metric-Aware Calibration:** Post-hoc bias calibration directly optimizes the competition asymmetric metric rather than standard accuracy, resulting in consistent improvement (~0.19 points) across all folds.
- **Ensemble Averaging:** Combining logit scores from 3 independently trained models reduces prediction variance and improves generalization, especially for rare classes.
- **Class Imbalance Handling:** WeightedRandomSampler ensures rare pathologies are adequately represented in each mini-batch, preventing the model from collapsing to No Finding for all predictions.

5.2 Limitations & Observations

- **Rare Class Collapse:** Despite WeightedRandomSampler, several rare classes received zero predictions in the final submission, indicating the model struggled to learn robust representations for very low-frequency classes.
- **Limited Training Duration:** Only 5 epochs per fold were trained due to time constraints on the Kaggle GPU. Extended training with a learning rate scheduler could improve convergence.
- **Score Headroom:** Validation scores in the range of -4.4 to -4.5 indicate substantial room for improvement, particularly in reducing false negatives.

5.3 Potential Improvements

- **Stronger Backbones:** Replacing DenseNet-121 with EfficientNet-B4, Vision Transformers (ViT), or Swin Transformers could provide stronger feature representations.
- **Asymmetric Loss Training:** Replacing cross-entropy with a loss function that directly minimizes the asymmetric clinical penalty could better align training objectives with the evaluation metric.
- **Extended Training:** Longer training (15-20 epochs) with cosine annealing or OneCycleLR scheduling typically yields better convergence in medical imaging tasks.
- **Test-Time Augmentation:** Applying test-time augmentation by averaging predictions across multiple augmented views of each test image can reduce prediction noise.
- **Data Augmentation for Rare Classes:** Techniques like class-conditional oversampling or synthetic data generation for rare pathologies could improve coverage of low-frequency classes.

6. Conclusion

This project developed a deep learning pipeline for automated chest X-ray pathology classification across 20 thoracic disease categories. Using a fine-tuned DenseNet-121 backbone, 3-fold stratified cross-validation, weighted random sampling for class imbalance, and a greedy per-class bias calibration step, the model achieved a mean tuned validation score of -4.47 on the competition asymmetric clinical metric.

The most significant methodological contribution is the post-training bias calibration step, which bridges the gap between standard cross-entropy training and the clinical cost function. This step consistently improved scores by approximately 0.19 points across all three folds, demonstrating that metric-aware post-processing is a practical and effective strategy in medical AI competitions.

While the model demonstrates reasonable performance for common pathologies (Effusion, Infiltration, Atelectasis, Pneumothorax), rare class prediction remains a challenge. Future work should explore asymmetric loss functions during training, stronger transformer-based backbones, and more sophisticated data augmentation strategies for rare categories.

This project provided valuable hands-on experience with the complete deep learning workflow for medical image analysis, including the unique challenges of clinical evaluation metrics, class imbalance, and the high stakes of diagnostic AI systems.

References

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